



Courtesy ISRO

Sky HIGH

It's not just about rippling rockets and swish satellites. India's space programme has transformed the nation's agriculture, communications and education.

■ By M.G. Radhakrishnan

Many villagers in the coconut palm-filled Thump, a sleepy fishing hamlet in Thiruvananthapuram, Kerala, vividly remember a khaki-clad attendant carrying 3-inch-thick "toy" rockets on a bicycle and a group of earnest-looking men assembling "something" (a payload) in a tiled shed. Things have come a long way since then, with India putting up its most glorious demonstration at Sriharikota on November 16, 2007, when the Indian Space Research Organisation (ISRO) tested cryogenic technology to power the Geosynchronous Satellite Launch Vehicle (GSLV), capable of putting four-tonne satellites in orbit. It was the successful demonstration of a technology denied to India by the US for 10 years and as ISRO chairman, G. Madhavan Nair, put it, "The ultimate in the state-of-the-art rocket technology and the climax in India's self-reliance."

The iconic images—the bullock cart and the bicycle—of India's early space research are a thing of the past as the country now makes an annual revenue of more than \$7 million (Rs 28 crore) by marketing space services and hardware to Russia and Europe. It's a far cry from the early '60s when Vikram Sarabhai, architect of India's space programme, had said, "We don't have the fantasy of competing with economically advanced nations in the exploration of the

LAUNCH PAD: GSLV ready to lift off at Sriharikota

PRAMOD KALE, INSAT project director, 1986

"That bird (INSAT 1C) is vital and till we send it up we are all spending sleepless nights."